Dilawar Singh

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Founding / Senior Backend Engineer with 10+ years of experience building high-performance APIs, SaaS platforms, and data pipelines in Rust, Python, and C++. Proven record of taking products from PoC to production: designing scalable microservice architectures, operating self-hosted infrastructure, and shipping reliable cross-platform systems. Enthusiastic contributor to Free and Open Source Software (FOSS).

Skills

- **Languages:** Rust, Python, C, C++ (expert); Java, TypeScript, JavaScript, PHP (intermediate)
- **Backend & SaaS:** Microservices and distributed systems; high-performance gRPC/REST APIs (Axum, Actix, FastAPI, CodeIgniter, Laravel); API design and versioning; authentication and authorization
- **Data:** SQL, PostgreSQL, DuckDB; job queues; real-time event pipelines (~10,000 events/session); data parsing, validation, and lifecycle management
- **Observability & Infrastructure:** OpenTelemetry, Grafana; structured logging; incident response tooling; Docker, QEMU; CI/CD (GitLab, Jenkins); Ansible, Pulumi; self-hosted infrastructure management
- **Systems:** Linux kernel modules, eBPF, Windows WDK, Win32/POSIX APIs, IPC, SDK development (Rust, C++)
- **Other:** Systems architecture, parser development, event-driven systems, technical literature review, team leadership and hiring

Work Experience

Independent Consultant — Self-Employed, Bengaluru Mar 2026 – Present

- Designing and overseeing development of high-performance APIs (Rust, AMQP) for a cyber lab running near-real-time experiments on highly correlated, diverse datasets (audio, video, sensor events)
- Designing and developing scalable microservices supporting a virtual assistant platform for an automotive control system (Rust, C, Qualcomm SoC)

Senior Engineer — Dognosis, Bengaluru Jan 2025 – Feb 2026

- Aggregated ~30 GB/session and ~10,000 events/session of live sensor data into a single operational control plane used for real-time decision-making and incident response
- Set up and operated self-hosted infrastructure (GitLab, PostgreSQL, Grafana, CI/CD runners), including access control, backups, and deployment policies
- Established risk-focused planning and sprint rituals that reduced invisible work, clarified ownership, and gave leadership visibility into team throughput
- Built hiring pipelines and hired 5 engineers, scaling the team from 6 to 11 in 3 months

Cofounder & CTO — Subconscious Compute, Bengaluru Dec 2019 – Dec 2024

- Hired and led a team of 15 engineers to build Shepherd, an automated endpoint compliance and security SaaS platform

- Designed the core backend architecture (Rust/C++) for a cross-platform threat detection and mitigation engine; drove the platform from PoC to production across customer endpoints
- Authored technical memos and pitch decks to secure accelerator placements and support fundraising

Research Fellow & Google Summer of Code (GSoC) Mentor 2016 – 2018

- Wrote Python bindings (pybind11) and created packages for MOOSE and Smoldyn, significantly improving usability and adoption by the scientific community
- Added high-performance numerical solvers (C++, Boost, GNU Scientific Library) to the MOOSE simulator
- Mentored multiple Google Summer of Code (GSoC) contributors on GPU/CUDA-based acceleration over 4 years

Firmware Engineer — Kritikal Solutions, Noida Jul 2009 – Jun 2010

- Implemented a Kalman-filter-based image stabilization system for professional movie cameras (C++, multi-threaded RTOS)

Projects

- Built Hippo, an automated scheduling system for NCBS, eliminating thousands of manual emails and calendar conflicts (PHP, Vue.js)
- Published Rust crates: simple_accumulator (online statistics), query-wmi (Windows WMI interface), job-dispatcher (async job execution)
- Published Python packages: PlotDigitizer (extract raw data from plot images), SerialScope (serial-port oscilloscope)

Education

- **PhD, Computational Neuroscience** — NCBS Bangalore (TIFR Mumbai), 2019
- **PhD, Digital Systems** (withdrawn) — IIT Bombay, 2011–2013
- **M.Tech, Electrical Engineering (VLSI)** — IIT Bombay, 2009
- **B.Tech, Instrumentation & Control Engineering** — Dr. MGR ERI Chennai, 2007

Publications

Full list on Google Scholar

- Subunit exchange enhances information retention by CaMKII in dendritic spines, *eLife*, 2018
- Python interfaces for the Smoldyn simulator, *Bioinformatics*, 2022
- BioSimulators: a central registry of simulation engines, *Nucleic Acids Research (NAR)*, 2022